One protein, by condition

The intensities of a single feature, condition by condition.

AT A GLANCE

Where	Single protein ⟶ Boxplot
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation, and one feature selected. Conditions come from the annotation.
Data source	Data, Filtered Data or Imputed Data; the page opens on the most processed one that exists.
One box	The replicates of the selected feature within one condition — one value per sample, not one per feature.
Table	Yes, below the plot, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

This is the plot drawn after the volcano: not where a feature's point landed, but what the feature actually did. Every box is one condition and what fills it are that condition's replicates, so the question it answers is whether the difference a test reported is carried by all of them or by one.

A box here holds as many values as the condition has samples, which makes it a far smaller summary than the boxes in quality control, where a box holds one value per feature. Read the spread, not the exact whisker.

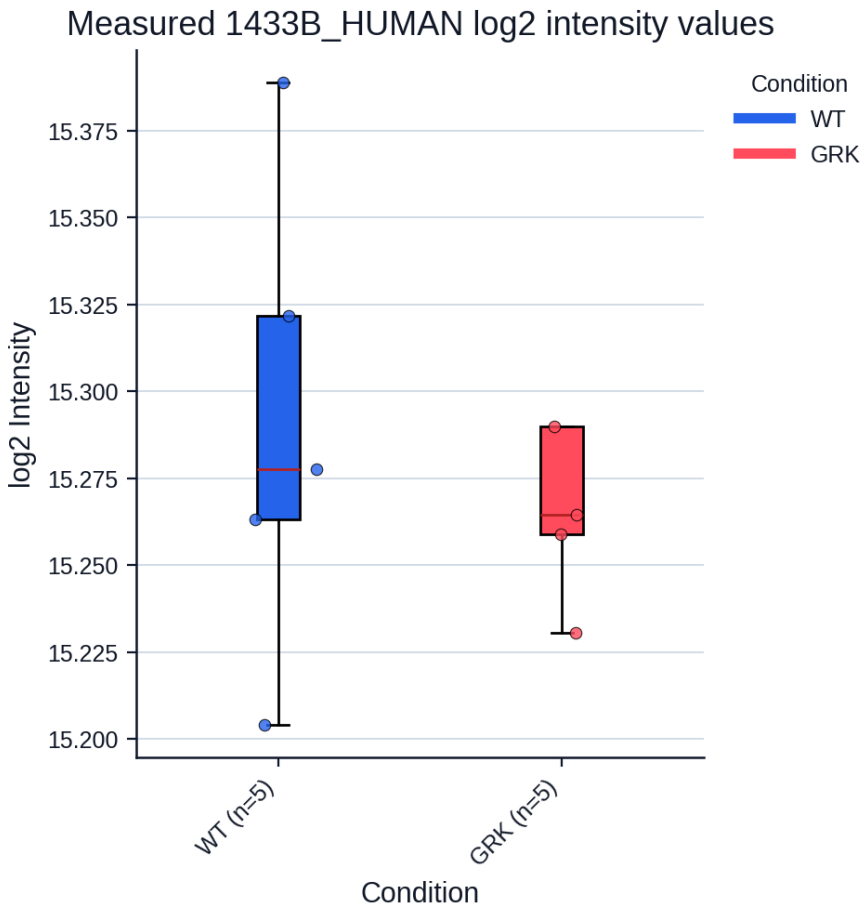


Figure 1.1. The protein 1433B_HUMAN on the example dataset, with **Toggle Dots** on: one box per condition, each built from the five replicates of that condition and labelled (n=5) underneath. The whole y axis spans under two tenths of a $\log_2$ unit, so the two boxes overlap almost completely — this is a feature that did not move, and the figure says so at a glance.

Reading it

- One box, one condition: the distribution of that feature's values over the samples of the condition.
- The line across it: the median of those values.
- The dots over the box, when **Toggle Dots** is on: the replicates that fall inside the whiskers, spread sideways so that equal values do not hide one another.
- The markers beyond the whiskers, when **Toggle Outliers** is on: the replicates the dots leave out. One of these in a set of five is a sample to look at, not a result.
- The label under each box: the condition, followed by (n=...) — the number of values the box was built from — as long as **Toggle Sample n** is on.

> EASY TO GET WRONG
>
> Three points are not a distribution. With three or four replicates the median is one value or the mean of two and a quartile is an interpolation between neighbours, so the box draws those few numbers rather than estimating a population. Compare the dots across conditions before the boxes, and treat a difference that rests on one replicate as one that rests on one replicate.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset the feature is taken from: Protein, Phospho or Phosphoprotein. Only the levels actually uploaded are offered; phosphoprotein appears once a collapse has been run.	Protein	page
Data source	Which matrix the values are read from: Data, Filtered Data or Imputed Data. Only the ones that exist are offered.	the most processed matrix available	page
Identifier	Which name the feature is chosen by: Protein Names, Phosphosite Names or Phosphoprotein Names according to the level, and Gene Names where the dataset carries a gene column. The phosphosite level offers no gene option.	Protein Names	page

Controls on this page (continued)

Control	What it does	Default	Where
Protein / Site	The one feature to draw; at protein level only the first accession of a group is offered. Labelled **Gene** under Gene Names.	the first entry	page
Conditions	Which conditions to draw. Their order is the annotation's.	all	page
Toggle Dots	Draws the individual replicates over each box, the outliers excluded.	off	page
Toggle Sample n	Appends (n=...) to each condition label — how many values that box was built from.	on	page
Toggle Outliers	Draws the values beyond the whiskers as individual markers.	off	page
Random Seed	Appears only with **Toggle Dots** on. Fixes the sideways jitter of the replicate dots, so that the same figure is reproduced exactly.	69	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm, DPI between 72 and 600.	20, 10, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Show title	The plot's own title is Measured <feature> log2 intensity values, and Imputed ... on Imputed Data. Its text, size, weight, colour and position are set here too.	shown	appearance
Show legend	The legend of condition colours, and its title, size, colour and frame.	shown	appearance
Box border color, Median color	The outline of every box, and the line across it.	black, #b22222	appearance
Data dot size, opacity, outline, outline width	The replicate dots drawn by **Toggle Dots**.	4.5, 0.8, black, 0.4	appearance
Outlier dot color, outline color, size	The markers drawn by **Toggle Outliers**, styled separately because they are the values the replicate dots exclude.	red, black, 5	appearance
Condition markers	The same four dot settings, one condition at a time, over the general ones above.	the general settings	appearance

In the plot appearance panel (continued)

Control	What it does	Default	Where
Condition names and colours	Renames a condition on the axis and in the legend, and recolours it. The colour is shared with every other plot in the session.	the condition palette	appearance
Frame, grid, font, background	Which of the four spines are drawn and the grid behind the boxes; one face for the whole figure or one per text; paper colour, transparency and the preset themes.	all four spines, no grid	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here. **Toggle Header** and **Toggle Legend** are no longer beside the plot: they are **Show title** and **Show legend** in the panel.

3 How it is computed

The box, the whiskers and what counts as an outlier follow the standard definition given in **Concepts**; it is not repeated here. What is particular to this page is only what goes into the box: the replicates of one feature within one condition, one value per sample.

HOW IT IS COMPUTED

The rows of the chosen matrix whose identifier column matches the selection are taken: at protein level on the first accession of the group, by gene on any of a row's semicolon-separated gene tokens, so more than one row can match.
For every sample column j the value plotted is the median over the matching rows,

$$v_j \;=\; \text{median}\{\, x_{ij} \;:\; i \in M \,\}, \qquad M = \text{the matching rows},$$

computed over the rows that carry a value in that sample. With one matching row — the ordinary case — v_j is that row's value. Non-numeric entries, $\pm\infty$ and NA are removed first, and a sample left with nothing is dropped rather than counted as zero. The box of condition c is built from $\{\, v_j \;:\; j \in S_c \,\}$, the values of the samples of that condition, so its height is the spread *between replicates* and n is at most the number of samples in the condition.

WHICH CONDITIONS GET A BOX

A box is drawn for a condition with **three or more** values. A condition below that is not dropped: it keeps its place on the axis, its label, its colour in the legend and its row in the table, and is drawn as its individual points instead — which appear whether or not **Toggle Dots** is on. Only a condition in which the feature has *no* finite value leaves the axis entirely.

THE REPLICATE DOTS

With **Toggle Dots** on, the values within $Q_1 - 1.5\times\text{IQR}$ and $Q_3 + 1.5\times\text{IQR}$ are drawn over the box, each displaced sideways by a draw from a normal distribution of standard deviation 0.05 around the box's position. The generator is seeded with **Random Seed**, so one seed reproduces one scatter exactly.

Edge cases

- A missing value removes that sample from the box; it is not imputed here and does not become a zero. The *n* in the label is what survived, which is why it can differ between conditions of equal size. A feature missing throughout a condition leaves that condition off the plot and out of the table, and if that is true everywhere the figure reads No measured values in the selected conditions.
- When the selection matches several rows — duplicates, or a gene mapping to several proteins — every plotted value is their per-sample median, and an amber note above the table says how many rows were combined.
- On Imputed Data some values in the box are draws from the imputation model rather than measurements, which is why the title then reads Imputed instead of Measured: the exported figure carries no data-source selector.
- At phosphoprotein level collapsed with robust_site_median the values are ratios to each site's own across-sample median, and the axis says log2 Ratio to Site Median instead.

The threshold of three is fixed and there is no control for it: below three values a box would describe nothing, so the replicates are shown as points instead.

4 The table below the plot

One row per condition that reached the plot, in the order the conditions were drawn.

Columns in this table

Column	What it holds
Condition	The condition, as labelled on the axis.
N, SampleCount	How many values the box was built from — the same number as (n=…) in the label, twice, under the two names the single-protein tables use.
Mean, Median	Mean and median of those values, to four decimals. The median is the line across the box.
Q1, Q3	The 25th and 75th percentiles: the bottom and top of the box.
Min, Max	The smallest and largest value, outliers included — the data, not the whisker caps.
SourceFeatureCount, SourceFeatures	How many rows the selection matched and which — 1 unless a gene maps to several features or the matrix holds duplicates.
Aggregation, Warning	Per-sample median, and a note filled in only when more than one row was combined. The note is shown above the table as well.

5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

The function looks the feature up in the frame's index, so the frame has to be indexed by the identifier first:

PYTHON

```
data_by_protein = data.set_index("ProteinNames", drop=False)
fig = boxplot_int_single_prot(data_by_protein, meta, "1433B_HUMAN", dots=True)
```

R

```
data_by_protein <- data
rownames(data_by_protein) <- data_by_protein$ProteinNames
fig <- boxplot_int_single_prot(data_by_protein, meta, "1433B_HUMAN", dots = TRUE)
```

At phosphosite level the same call takes data3 indexed by PTM_Collapse_key and annotated with meta2. dots and outliers are **Toggle Dots** and **Toggle Outliers**, and the per-sample median over matching rows is the same aggregation the page performs — the function writes it into the title rather than into a warning above a table. Two things differ: the (n=…) in each label is always printed, there being no **Toggle Sample n**, and the boxes take their colours from Matplotlib's default cycle instead of the session's condition palette unless plot_colors is given.

6 Exporting

Width and height control the rendered figure; the browser then scales its preview to the available space. Resolution controls raster detail. Choose PDF for a figure that will be scaled or edited, PNG for a slide. The table downloads separately as CSV, and it is the better thing to keep: it holds the numbers the box is drawn from.

7 Further reading

NIST: box plots explains the summary geometry. CoPYlot's minimum-count rule and optional individual points are described above.

Behind this page

Function	What it produces
single_protein_boxplot_plot	The figure, with the condition palette and the appearance panel applied
single_protein_boxplot_table	The per-condition summary under it
boxplot_int_single_prot	The same plot in the package, for a notebook